

MTH601

Midterm important questions 2024

VU360

Lecture 1 – 5

- Vector addition: $a + b = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$
- Vector scalar multiplication: $ka = (ka_1, ka_2, \dots, ka_n)$
- Dot product: $a \cdot b = a_1b_1 + a_2b_2 + \dots + a_nb_n$
- Magnitude of a vector: $|a| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$
- Equation of a line: $(x, y) = (x_0, y_0) + t(a, b)$

Lecture 6 – 10

- Matrix addition: $A + B = [a_{ij} + b_{ij}]$
- Matrix multiplication: $AB = [\sum (a_{ij} * b_{kj})]$
- Inverse of a matrix: $A^{-1} = \frac{1}{\det(A)} * \text{adj}(A)$
- Determinant of a matrix: $\det(A) = a(ei - fh)$
- Cramer's rule: $x = \frac{\det(Ax)}{\det(A)}$

Lecture 11 – 15

- Linear transformation: $T(x) = Ax$
- Kernel and image of a matrix: $\ker(A) = \{x \mid Ax = 0\}, \text{im}(A) = \{Ax \mid x \in \mathbb{R}^n\}$
- Eigenvalues and eigenvectors: $Ax = \lambda x$
- Diagonalization of a matrix: $A = PDP^{-1}$
- Orthogonality and orthonormality: $u \cdot v = 0, |u| = 1$

Lecture 16 – 22

- Differentiation: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- Integration: $\int f(x)dx = F(x) + C$
- Double and triple integrals: $\int \int f(x, y)dx dy, \int \int \int f(x, y, z)dx dy dz$
- Vector calculus: $\nabla f, \nabla \cdot F, \nabla \times F$
- Laplace's equation: $\nabla^2 f = 0$